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Acta Tropica 89 (2004) 109–123
ACTA TROPICA
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Opportunities and limiting factors of intensive vegetable farming in malaria endemic Côte d'Ivoire

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Abstract

Poverty reduction policies guide development strategies. In economies that depend heavily on agriculture, in the face of rapid population growth, innovative approaches are required to satisfy food needs, increase household welfare and alleviate poverty. Irrigated agriculture is an important strategy to enhance crop production, but it must be well tailored to specific socio-ecological settings, as otherwise, it might increase the burden of water-related parasitic diseases and delay economic advance. The purpose of this study is to assess and quantify the effect of ill health, particularly malaria, on the performance of farm activity, with an emphasis on drip-irrigated vegetable farming in rural Côte d'Ivoire. Vegetable yields and revenues were monitored among 12 farmers and linked with longitudinal medical and entomological surveys. Over the course of 10 months, farmers were classified as sick, on average, for 14–15 days, with malaria accounting for 8–9 days (58%), confirming that malaria is the most important disease in this setting. There was a large heterogeneity among farmers, with malaria-related work losses ranging between 0 and 26 days. Work absenteeism correlated with overall yields and revenues. During a single cabbage production cycle, those farmers who were prescribed sick because of malaria for more than 2 days (mean: 4.2 days) had 47% lower yields and 53% lower revenues than farmers who missed a maximum of 2 days (mean: 0.3 days). This is consequential in an intensive cropping system, where substitutes for qualified workers are not readily available. We conclude that mitigating the burden of malaria is an important step towards reducing the vulnerability of people engaged in intensive agricultural production. This calls for targeted interventions to facilitate agriculture-based rural development that might spur social and economic development and reduce inequities in sub-Saharan Africa.
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Keywords: Agricultural intensification; Côte d'Ivoire; Drip irrigation; Labour force productivity; Malaria; Poverty reduction; Rural development; Vegetable farming

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doi:10.1016/j.actatropica.2003.08.004

1. Introduction

Over the past decade, the United Nations and other international organisations set forth a series of resolutions and agreements after the recognition that improvements in living conditions mainly occurred in industrialised countries, while progress in the developing world was uneven and slow. These findings were presented at major world conferences and culminated, in September 2000, in the articulation of the millennium development goals (MDG; http://www.developmentgoals.org/). The overarching objective of the MDG is to reduce poverty, and ambitious targets have been set for the year 2015. Coupled with the current evidence-base that improving the health conditions of the global poor could avert millions of premature deaths and mitigate the global burden of disease, these movements have the potential to substantially reduce inequities and facilitate unprecedented social and economic development (WHO, 2001; Jha et al., 2002; Victora et al., 2003).

Recent analyses, however, revealed that it might be difficult to achieve the MDG, particularly in sub-Saharan Africa (Sahn and Stifel, 2003). This region's economy continues to be heavily dependent on agriculture and increase in food production has been outpaced by population growth (http://apps.fao.org/page/collections?subset=agriculture). In view of the predicted demographic developments of the coming 25 years (United Nations, 2002), major challenges lay ahead for Africa's agriculture to meet food demands, and in turn make significant progress towards the MDG. At present, agriculture in Africa primarily depends on a human workforce that is hampered by poverty and ill health. In addition, the extension of the arable surface is threatened by desertification and depletion. Consequently, innovative approaches must be developed that simultaneously aim at economic growth, improvement in health status and social wellbeing, and the protection of the natural environment.

Irrigation is a prominent example that exhibits the intimate connection between agriculture, health, wellbeing and the environment. On the one hand, irrigation is a key strategy to enhance crop production systems (Verma, 1986), but on the other hand, it often results in negative health outcomes, which is consequential to the increased frequency and transmission dynamics of water-associated infectious diseases (e.g. schistosomiasis) or water-related vector-borne diseases (e.g. malaria) (Kitron, 1987; Hunter et al., 1993; Gomes, 1998; Jobin, 1999). While there is strong evidence for a positive association between crop irrigation and increased transmission dynamics of schistosomiasis, this issue is far more complex in the case of malaria. In many areas of sub-Saharan Africa where malaria transmission is stable, the introduction of irrigated agriculture has little impact on the disease. In fact, several studies have documented that there is less malaria in communities living in close proximity to irrigation schemes when compared to populations living further away, which is partially explained by enhanced income that facilitates better protective measures to be taken (Boudin et al., 1992; Carnevale et al., 1999; Ijumba and Lindsay, 2001; Ijumba et al., 2002). For a better understanding and to propose mitigation strategies, actions have to be taken at different levels and by a multiplicity of stakeholders (Zehnder et al., 2003). Sound grass-root experiences across diverse ecosystems will be essential to generate an unmatched evidence-base for subsequent policy recommendations at the next higher levels.

The rural development project presented here operates at the interface of agriculture, health and wealth. The purpose of this paper is to assess the feasibility and quantify the microeconomic consequences and health impact of a project that aims at agricultural intensification through off-season vegetable farming in a malaria endemic setting of central Côte d'Ivoire. Farmers were trained and equipped to operate a new drip irrigation scheme. Agricultural productivity was measured and linked with cross-sectional and longitudinal entomological, medical and socio-economic surveys, for monitoring risk factors associated with rural development. Emphasis was laid on the least controlled risk factor, namely the vulnerability of the labour force to ill health, particularly malaria.

2. Materials and methods

2.1. Study area and population

The study was carried out between December 1999 and August 2002 in the village of Tiémélékro, located in the district of Dimbokro, 83 km south-east of

Yamoussoukro, the capital of Côte d'Ivoire, in the savannah zone of the country (geographic co-ordinates: −4.167°W, 6.500°N). The area is characterised by a transitional equatorial climate. Long-term annual averages of temperature and precipitation, measured at Yamoussoukro, are 27 °C and 1152 mm, respectively. In the years 2000 and 2001, the mean annual precipitation, recorded in the study village, was 1160 mm. The rainfall pattern is bimodal with a long rainy season from March to June and a shorter one from August to October. In 1998, the estimated population of Tiémélékro was 14,138 (Institut National de Statistique, Côte d'Ivoire). There are five primary schools and a college, all of which lack water supply and sanitation. The health infrastructure consists of two local health centres operated by one physician and two nurses. A small dam was constructed in the village in 1975 with the intention to boost agricultural production and livestock rearing. This water source is also utilised for washing clothes and dishes, and for recreational activities. The principal health problems, reported and observed during an exploratory survey shortly before the project was launched, were malaria, diarrhoeal diseases (epidemics in 1997), and respiratory infections, and to a lesser extent also soil-transmitted helminthiasis and urinary schistosomiasis (E.K. N'Goran, unpublished data). Table 1 summarises the project evolution and key activities.

2.2. Project rationale

The rationale behind the project was that agriculture-based income might be increased through the implementation of small-scale intensive production of selected food crops. Vegetables are relatively expensive commodities during part of the year, and are characterised by a stable demand and generally an elevated risk for the producer. Vegetable farming is labour intensive and requires high investments, as these crops are susceptible to biotic and abiotic stresses that can severely impair quality and yield. Control measures are necessary to reduce these risk factors, so that good yields of high quality vegetables can be obtained. Preceding agricultural market surveys in Abidjan, the business centre of Côte d'Ivoire, exhibited strong temporal fluctuation of vegetable

Table 1
Implementation of drip irrigation for intensive off-season vegetable farming in the village of Tiémélékro, central Côte d'Ivoire: project evolution and key activities

Time point	Duration	Project activity
December 1999		Project start in study village
May 2000	6 days	Cross-sectional survey no. 1: appraisal of malaria parasite rate among school children
April 2001	3 days	Community-awareness campaign for promotion of insecticide-treated bednets
June 2001	1 day	Project farmers receive first-aid box, containing essential drugs for treatment of common ailments including anti-malarials
June 2001	1 year	Health record keeping of study farmers (admission date, medical diagnosis and treatment, number of days prescribed sick, success of treatment)
August 2001	1 week	Entomological survey no. 1: larval and adult mosquito catches, species identification, assessment of sporozoite, oocytes and EIR
September 2001	4 months	Vegetable farming: cycle 1 with okra
December 2001	2 months	In-depth socio-economic survey among 64 randomly selected households
January 2002	6 months	Vegetable farming: cycle 2 with cabbage
February 2002	1 week	Entomological survey no. 2: larval and adult mosquito catches, species identification, assessment of sporozoite, oocytes and EIR
April 2002	1 week	Entomological survey no. 3: larval and adult mosquito catches, species identification, assessment of sporozoite, oocytes and EIR
June 2002	6 days	Cross-sectional survey no. 2: appraisal of malaria parasite rate
June 2002	1 week	Entomological survey no. 4: larval and adult mosquito catches, species identification, assessment of sporozoite, oocytes and EIR
August 2002	1 week	Entomological survey no. 5: larval and adult mosquito catches, species identification, assessment of sporozoite, oocytes and EIR

prices. Low prices were recorded towards the end of the rainy season. This is explained by the abundance of local, mainly rain-fed vegetables. Prices increased significantly during the dry season when vegetables were imported from neighbouring countries, namely Burkina Faso and Mali, where irrigated agriculture is currently more widely employed and production is climatically less threatened by biotic stress (Dao et al., 2003). The underlying idea of the present project was that the implementation of a farm-level drip irrigation system would facilitate tapping of high off-season prices for vegetables, which in turn would compensate for the relatively high production costs.

At the beginning of the project, farmers' health and the general wellbeing of the community were identified as key factors for overall project acceptance. Fig. 1 depicts the conceptual framework of the project employing a system dynamics approach (Sterman, 2000). It also emphasises the complex interplay between agriculture, health and wealth, as illustrated by the qualitative model showing the principal reinforcing and balancing feedback loops.

2.3. Vegetable farming and productivity monitoring

After presenting the objectives and intended procedures of the study to the village authorities, namely the adoption of drip irrigation to facilitate intensive off-season vegetable production, the village authorities were invited to appoint farmers to participate in

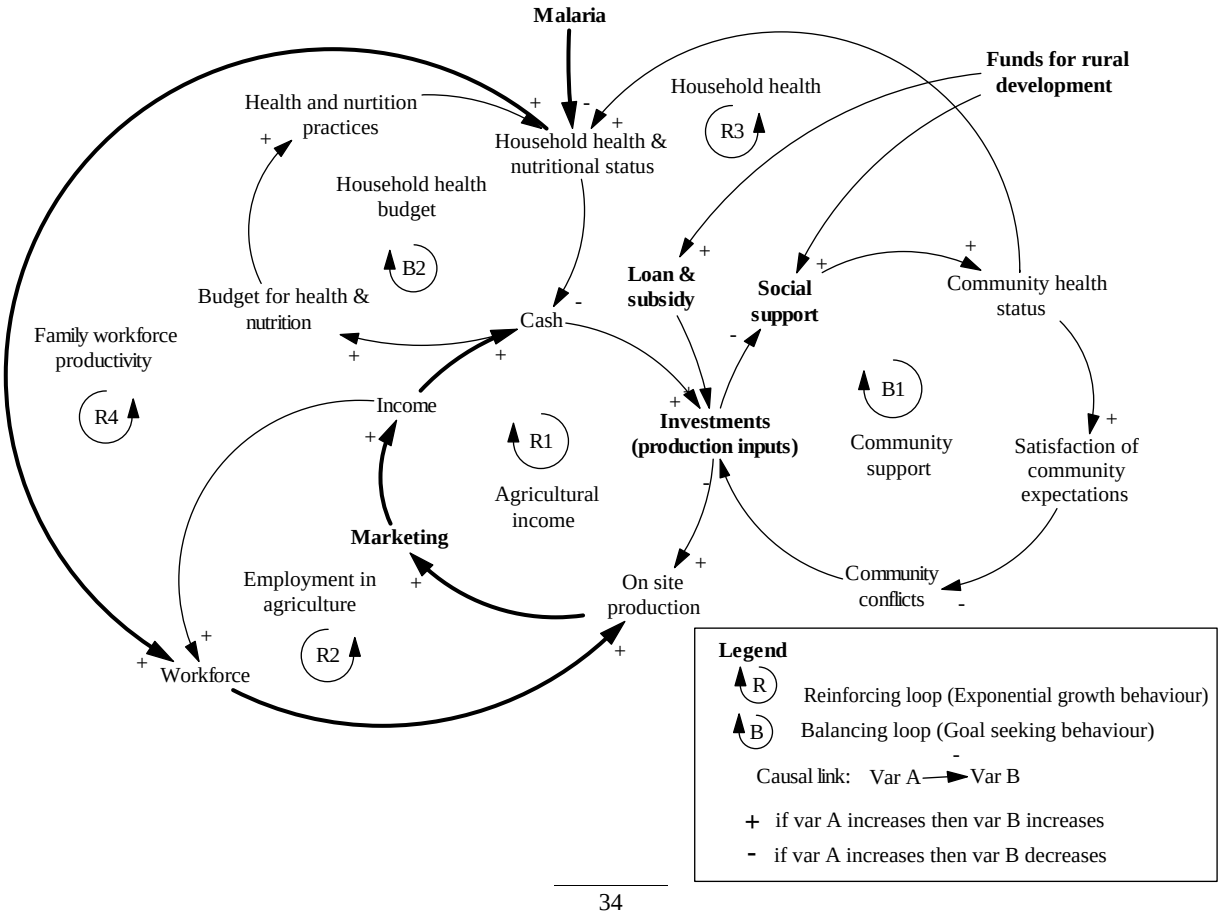


Fig. 1. Conceptual framework of intensive vegetable farming facilitated through drip irrigation and interplay between agriculture, health, and wealth emphasised by reinforcing and balancing feedback loops.

the study. Our research and development team had no control in this selection process. Twelve farmers agreed to participate. To enhance project acceptance before the actual implementation of the drip irrigation scheme, participating farmers received a first-aid box containing essential drugs for treatment of common ailments, including anti-malarial drugs. In turn, farmers were invited to visit the local health workers to report episodes of diseases and injuries, so that ill health could be monitored longitudinally. Forms were provided to systematically collect information pertaining to the admission date, medical diagnosis and treatment, the number of days a farmer was classified as sick, and whether symptoms disappeared thereafter.

Production of two different vegetable types with alternate cycles was launched in September 2001 (Table 1). It consisted of okra (*Abelmoschus esculentus*) in the first 4 months and cabbage (*Brassica oleracea* L.) in the next 6 months (the cycle for cabbage is only 4 months, but since not all farmers had synchronous itineraries it lasted up to 6 months). During the okra cycle, three producer groups designated A, B and C, were formed by randomly assigning four farmers to each group. The main reason for this strategy was that the initial irrigation system was centralised and not designed for individual production. During the subsequent cabbage cycle, the production was completely individualised, employing a family-level drip irrigation system. It consisted of either one or two irrigation units per farmer, with allocations according to the farmer's overall working capacity. This was estimated by the extent of other current agricultural activities. Each irrigation unit employed a 1000 l tank for approximately 500 m^2 of irrigated surface. Water was supplied by a diesel-driven pump, which was connected to all 12 plots. The initial capital investment per irrigation system was approximately US$ 500, offered as interest-free credits reimbursable over 3 years.

Besides irrigation, improved plant genotypes, mineral fertilisers, and repeated phyto-sanitary treatments were part of the technical itinerary. Per production cycle these variable costs amounted to US$ 30–50 per farmer and were offered as interest-free credits, reimbursable over a single production cycle. A technician was trained and then designated to the study site for monitoring all agricultural itineraries and providing advice when required. Marketed vegetable yields were recorded at sale expressed as kg/m^2 of irrigated surface. Since all production factors except work and land were pre-financed, and vegetable marketing was carried out with the assistance of the technician, profitability of labour and land was readily quantifiable, using the following formula: gross benefit = gross margin − costs of inputs.

2.4. Surveys

In April 2001, a community-awareness campaign was carried out to promote the use of insecticide-treated bednets for prevention of malaria. Between December 2001 and February 2002, an in-depth socio-economic survey was carried out in 64 randomly selected households. It comprised data collection on demographics (number, age and sex of individuals living in each household), professional and economic activities of the adult population, assessment of construction material for walls and roofing of houses, connection to the power grid, and appraisal of wealth indicators based on the presence of various household assets (e.g. bicycle, radio, etc.). Household heads were interviewed about major health problems, preventive measures against mosquito bites (e.g. use of bednets, fumigating coils, etc.), the availability of latrines, and disposal of wastewater and solid waste.

Entomological surveys were conducted every other month for 1 year commencing in August 2001. Due to unstable socio-political conditions and logistical difficulties, two of the surveys had to be cancelled. Adult mosquitoes were collected with two different methods, namely (i) human night bait catches (eight persons for three nights per survey), and (ii) spray catching, employing insecticides inside five sentinel houses early in the mornings (two nights per survey). In addition, mosquito larvae were systematically collected in four pools with standing water by two experienced field workers, dipping with small kitchen sieves mounted on 2 m long wooden sticks for 15 min at each site. Adult mosquitoes and larvae were brought to the laboratory and identified at genus and species level. The physiological age, and hence parity, of adult female *Anopheles* was determined by dissection of ovaries and examination of tracheoles (Detinova, 1962). The identification of species of sporozoites and oocysts obtained from dissected specimens was carried out by enzyme-linked immunosorbent assay (ELISA), according to standard procedures (Beier

et al., 1988). Furthermore, stomach content analyses of adult *Anopheles* were done to estimate the proportion of blood meals taken from human hosts. Antibodies for the tropical preference of vectors were chosen with respect to abundance of alternative hosts in the study area, namely anti-human, anti-bovine, anti-porcine and anti-chicken antibodies, previously marked with peroxidase. The entomological inoculation rate (EIR), i.e. the number of infective bites per person per unit of time, was also estimated, using the landing catch data only (MacDonald, 1957).

Two cross-sectional surveys, covering a period of 6 days each, were carried out in the study village to assess malaria parasite rates, primarily among school-age children. The first survey was done shortly after the beginning of the project in May 2000. A total of 389 school children, aged 7–11 years, were examined. The second survey was carried out in June 2002 and enrolled 118 school children in the same age group, as well as 87 children aged 2–6 years, and 68 individuals aged 12 years and above. Finger prick blood samples were taken from all participants and thick and thin blood smears were prepared on microscope slides. After the slides were stained with Giemsa, they were microscopically examined. Parasite identification (*Plasmodium falciparum*, *Plasmodium malariae* and *Plasmodium ovale*) and appraisal of species-specific parasitic load were determined in the nearby laboratory by the medical personnel of the national malaria control programme of Côte d'Ivoire, following standard, quality-controlled procedures.

2.5. Routinely collected disease records and statistical analysis

Disease records, routinely collected by the personnel in the two health centres of the study village, were obtained from January 1997 to December 2001. All cases of malaria, respiratory infections and diarrhoeal diseases were counted and expressed as the monthly number of cases per 1000 inhabitants. Correlation coefficients between the different diseases were computed on version 7.0 of the STATA software package (Stata Corporation, College Station, USA).

Data from the household survey were entered and analysed on version 6.04 of the EpiInfo software package (Centres for Disease Control and Prevention, Atlanta, USA). For those households with complete data records, the frequencies of different house construction materials, connection to the power grid, different household assets and protective measures against mosquito bites were computed. Multivariate analysis, employing logistic regression modelling, was utilised for appraisal of significant associations between the presence of any protective measure against mosquito bites (dependent variable) and house construction material, connection to the power grid, possession of bicycle, motorcycle, radio, television and bednet, and the perceived significance of malaria. A baseline model was established with those households utilising any protective measure against mosquito bites being defined as cases. Non-significant associations ($P > 0.05$) were removed one at a time by a backward elimination technique. For those associations that remained significant, the adjusted odds ratio (OR), including 95% confidence intervals (CI), the likelihood ratio statistics and the corresponding *P*-values were calculated. The same approach was utilised for assessment of significant associations between the above-mentioned variables and the possession of a bednet and malaria being perceived as a major health problem.

Finally, the relationships between individual farmer's vegetable yields during the cabbage cycle and the number of days classified as sick due to diseases or injuries, or malaria, were investigated. Farmers were stratified into two groups according to the total number of days that they were classified as sick due to malaria for comparison of gross margins and revenues, using simple ANOVA analyses.

3. Results

3.1. Reported ill health in Tiémélékro

Routinely collected disease records at the health care facilities of Tiémélékro revealed that malaria, respiratory infections, and diarrhoeal diseases are the main reasons for visiting a healthcare professional, together accounting for 99% of the total visits. The number of monthly cases per 1000 inhabitants for these three causes of ill health is depicted in Fig. 2. Malaria was the single most important cause for seeking health care. Over the past 5 years, this disease accounted for 55% of the total consultations with a mean annual in-

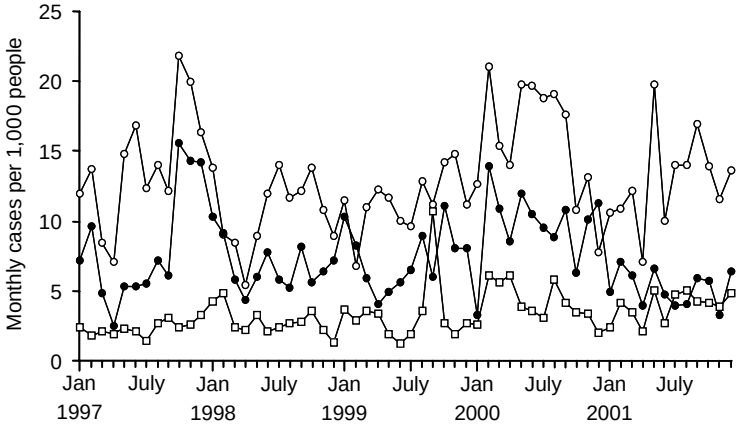


Fig. 2. Monthly number of cases diagnosed with malaria (○), respiratory infections (●) and diarrhoeal diseases (□) per 1000 people presenting at the health care facilities of Tiémélékro from January 1997 to December 2001.

cidence of 155 clinically diagnosed cases per 1000 inhabitants. Seasonal patterns, as expected by variation in precipitation and ambient temperature were not clearly visible in the time-series of clinically diagnosed malaria cases, indicating that there is stable and intense malaria transmission in this area. While a relatively high correlation coefficient was observed between malaria and respiratory infections ($r = 0.58$), the correlations between malaria and diarrhoea and between respiratory infections and diarrhoea were weak, namely 0.30 and 0.13, respectively.

The remaining 1% of recorded diseases included allergic conjunctivitis, measles, *Mycobacterium ulcerans* disease (Buruli ulcer) and urinary schistosomiasis. A few cases of onchocerciasis were also reported.

3.2. Economic activities, household characteristics and protection against malaria

Complete data records were obtained from 63 households. The interviews with the household heads revealed that the large majority of the adult population is engaged in subsistence agriculture (82%). Rice and maize are the two predominant food crops. There is some cash cropping of coffee and cocoa, but plantations are old and of low productivity. Farmers adhere to these crops on traditional grounds and for maintaining a minimal source of income. An emerging cash crop appears to be rubber, as a rapidly increasing number of seedling nurseries are seen in the region. Among female respondents only, 9% responded that their principal activity was housework. The percentages of small traders and craftsmen were very small. The lack of a sound transport infrastructure is an important reason why trading currently is of marginal economic significance in this setting.

Corrugated iron-sheets and roughcast mud are the predominant construction materials for roofing and house walls, accounting for 94 and 58%, respectively. Conversely, the use of cement bricks for wall construction and traditional thatched-grass roof types were recorded in 42 and 6% of the surveyed houses, respectively. Connection to the power grid was high since 70% of the households had electricity. The majority of households (82%) disposed of their wastewater and solid wastes near their homes, and only a few of the surveyed households (17%) had some kind of sanitary facilities for excreta disposal. Forty-tree households (68%) had at least one radio and 20 households (32%) reported possession of a television. Bicycles and motorbikes were recorded among 27 (43%) and 6 (10%) of the households, respectively.

Overall, 33 of the households (52%) reported no particular means of protection against mosquito bites. The remaining 30 households reported the use of bednets ($n = 11$, 18%), fumigating coils ($n = 9$, 11%), insecticide sprays ($n = 5$, 9%), traditional medicine ($n = 4$, 6%) and modern anti-malarial drugs ($n = 1$, 2%) as their principal means of protection. Table 2 shows the results from the multivariate analyses.

Table 2
Multivariate analysis employing logistic regression modelling for appraisal of different household variables and perceived malaria importance that are significantly associated with three different outcome variables ($n = 63$ households)

Variable	Adjusted odds ratio (95% CI)	Likelihood ratio statistics	*P*-value
Protective measures against mosquito bites			
Malaria perceived as health problem	4.40 (1.29–14.92)	6.28	0.012
Connection to power grid	3.70 (0.96–14.29)	3.95	0.047
Iron-corrugated roof	0.07 (0.04–1.04)	4.46	0.035
Possession of bednet			
Connection to power grid	6.55 (1.22–35.07)	6.18	0.013
Possession of bicycle	0.15 (0.04–0.63)	8.08	0.005
Malaria perceived as health problem			
Possession of bednet	4.75 (0.97–23.34)	4.72	0.030

Households that employed any protective measure against mosquito bites were significantly more likely to perceive malaria as a major health problem (adjusted OR = 4.40, $P = 0.012$), and to be connected to the power grid (adjusted OR = 3.70, $P = 0.047$), but were less likely to have iron-corrugated roofing (adjusted OR = 0.07, $P = 0.035$) when compared to those households using no protective measures against mosquito bites. Possession of a bednet was positively associated with connection to the power grid (adjusted OR = 6.55, $P = 0.013$) but was negatively associated with the possession of a bicycle (adjusted OR = 0.15, $P = 0.005$). Finally, households that perceived malaria as a major health problem were significantly more likely to have a bednet than the other households (adjusted OR = 4.75, $P = 0.030$).

3.3. Entomological findings

Anopheles gambiae sensu lato and *Anopheles funestus* were identified as the two principal malaria vectors. Assessment of abundance of adult mosquitoes revealed that the former species is the predominant one, which was also confirmed by the number of larvae collected in the surveyed pools. Both species showed strong temporal variation in total abundance, which is expected by the seasonal climatic variations. The entomological surveys revealed that for *A. gambiae s.l.* a low biting rate of 1.25 per person per night was found in February, whereas a high rate of 36.3 was recorded after the rainy season ceased (Fig. 3a). Overall, 3.2% of the dissected mosquitoes were infectious, thus it is estimated that the mean EIR is 291 infectious bites per person per year. The highest EIR, namely 2.3 infectious bites per person per night, was observed in August 2001 and the lowest, approaching 0, in February 2002 (Fig. 3b).

A high positive correlation of 0.89 was obtained between the parity of *A. gambiae s.l.* and the infection rate. The percentage of female mosquitoes of potentially infectious physiological age was generally above 60%, but in April 2002 it was considerably lower. Detailed analyses of the stomach content revealed that the captured mosquitoes were 100% anthropophilic.

3.4. Malaria parasite rates

The two cross-sectional surveys revealed malaria parasite rates of *Plasmodium* sp. among school children aged 7–11 years of 74% in May 2000 and of 86% in June 2002, respectively. In the latter survey younger and older individual were also screened; children aged 2–6 years showed a parasite rate of 80%, while individuals aged 12 years and above had a parasite rate of 68%. *P. falciparum* was the predominant species, accounting for 93 and 97% of all cases in the two surveys, respectively. *P. malariae* was identified in the remaining cases, while no single case of *P. ovale* was discovered.

3.5. Agriculture-health linkages

Detailed monitoring of the study farmers' health conditions over the 10 months of intensive vegetable farming clearly revealed that malaria was the pri-

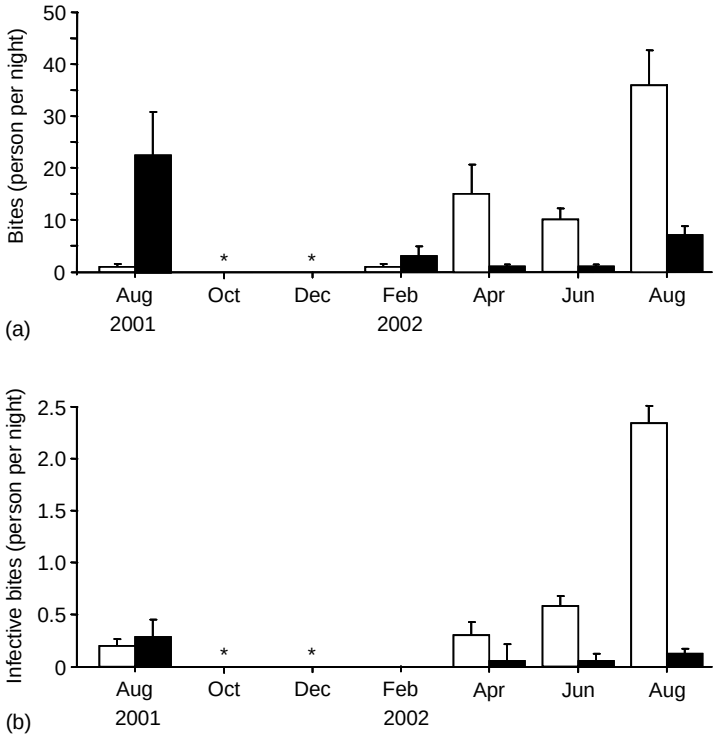


Fig. 3. Number of mosquito bites per person per night (a) and entomological inoculation rate (EIR) in Tiémélékro from August 2001 to August 2002 (b) (white bars: *A. gambiae s.l.*; grey bars: *A. funestus*; (∗) missing surveys due to unstable socio-political conditions).

mary reason for advising the farmers to stay at home (Table 3). All but 1 of the farmers were classified as sick for at least 1 day due to malaria and, on average, each farmer was classified as sick for 8–9 days because of malaria. There was a large heterogeneity among farmers with a maximum of 26 days classified as sick due to malaria. Three farmers were prescribed sick for 18–26 days within the study period, which translates to 6–9% of their potential work time lost on the farms. Out of 3600 days at risk of disease or injury, malaria accounted for 99 days, which corresponds to a daily risk ratio of 0.028. All other causes

Table 3
Reported ill health and number of days each farmer was classified as sick within a period of 10 months, commencing in September 2001

Health problem	Farmer												Total (%)
	A1	A2	A3	A4	B1	B2	B3	B4	C1	C2	C3	C4	
Malaria	26	20	18	3	10	6	3	1	7	3	2	–	99 (58)
Tiredness	6	8	8	3	3	–	–	–	–	–	3	2	33 (19)
Abdominal pain	–	1	–	–	6	8	–	3	–	–	1	–	19 (11)
Injury	–	–	–	–	–	–	–	12	3	–	–	–	15 (9)
Headache	–	–	–	–	–	–	–	–	3	–	–	–	3 (2)
Muscle ache	–	–	–	–	–	–	–	–	–	2	–	–	2 (1)
Total	32	29	26	6	19	14	3	16	13	5	6	2	171 (100)
Discontinuing with vegetable farming[a]	×	×	×	×	×			×					

[a] Farmer A4 ceased vegetable farming because of an hernia operation and farmer B4 started another business after he had gained enough money (shop owner). High likelihood that farmers A1, A2, A3 and B1 discontinued with vegetable farming because of ill health.

of ill health accounted for a cumulative average number of 6 days prescribed sick within the 10 months of observation. Overall, malaria accounted for 58% of all the time that the health personnel advised farmers to stay at home due to disease or injury. Tiredness, probably due to the hardship of manual field-work, was responsible for almost one-fifth of the prescribed duration of convalescence. This symptom could, at least partially, also be due to malaria. Six farmers discontinued intensive vegetable farming, four of whom were classified as sick for 19–32 days, with malaria accounting for 10–26 days.

The mean marketed okra yields by the three producer groups ranged between 0.35 and 0.74 kg/m^2 of irrigated surface. Drip irrigation was individualised for the subsequent cabbage cycle and the marketed yields exhibited very strong variation, from as low as 0.05 up to 1.19 kg/m^2 of irrigated surface. Stratification of farmers into 2 groups, namely (i) >2 days classified as sick due to malaria, and (ii) ≤2 days, showed that marketed cabbage yields and gross margins were 47 and 53% lower, respectively, in the highly malaria-affected group of farmers. ANOVA analyses between these two groups showed that gross margins were significantly higher among the less malaria-affected farmers ($P = 0.03$). Overall, 42% of the observed variation could be attributed to the effect of malaria (Table 4). Although the total marketed cabbage yield showed no statistical significance ($P = 0.06$), this might be simply explained by the small sample size of the two groups of farmers ($n = 6$ in each group). Stratification according to the total number of days farmers were classified as sick due to any disease or injury revealed similar results as in the case of malaria only. ANOVA analyses found that yields and revenues were significantly lower among those six farmers classified as sick more frequently than the second group ($P = 0.04$ and 0.02, respectively).

Table 4
Analysis of variance of marketed yield of cabbage and gross margin (expressed in FCFA) among 12 study farmers, stratified by the numbers days prescribed sick due to malaria during a single cabbage production cycle between January and June 2002

Number of farmers	No. of days classified as sick due to malaria	Marketed yield (kg/m^2)	Gross margin (FCFA)[a]
6	≤2 (mean: 0.3)	0.604	39982
6	>2 (mean: 4.2)	0.323	18775
$P > F$[b]		0.06	0.03
R^2-value (%)		36	42

[a] US$ 1 corresponds to FCFA 656 (as of early 2003).
[b] Level of significance between the two groups.

Finally, Fig. 4a and b show the mean marketed yields of okra and cabbage for the three producer groups A, B and C in relation to the mean number of days prescribed sick due to malaria. Visual examinations reveal negative correlations between total yield and the number of days prescribed sick due to malaria, analogous to the individual farm-level observations during the single cabbage cycle.

4. Discussion

Poverty reduction is the ultimate aim of the MDG, and this issue is of central importance to tailor and implement national development policies, particularly in the developing world. Efforts are underway, and must be further strengthened, to monitor closely the social and economic conditions of populations, including equity, and to assess the impact, efficiency and cost-effectiveness of poverty-oriented policies, programmes and projects (Ellis and Bahiigwa, 2003; Mukherjee and Benson, 2003; Sahn and Stifel, 2003; Victora et al., 2003). An important issue that needs to be addressed is how progress towards poverty reduction can be made in settings that primarily depend on subsistence agriculture and are characterised by continuous high population growth rates, e.g. sub-Saharan Africa (United Nations, 2002).

Our study was an attempt to assess and quantify this issue at the local level, within the frame of a small-scale project, carried out in a rural part of the savannah zone of Côte d'Ivoire. We have shown that the strategy of producing high quality off-season vegetables was feasible and might be profitable. The increase in agricultural income was initiated through the implementation of a drip irrigation system and farmers' encouragement to engage in the cultivation and marketing of risk-diversifying crops, namely cabbage and okra. However, launching of the project necessitated important financial investments, and required a considerable amount of external advice and support for knowledge and technology transfer. Farming of cabbage, in our case, not only required ordinary agricultural operations such as field preparation, sowing,

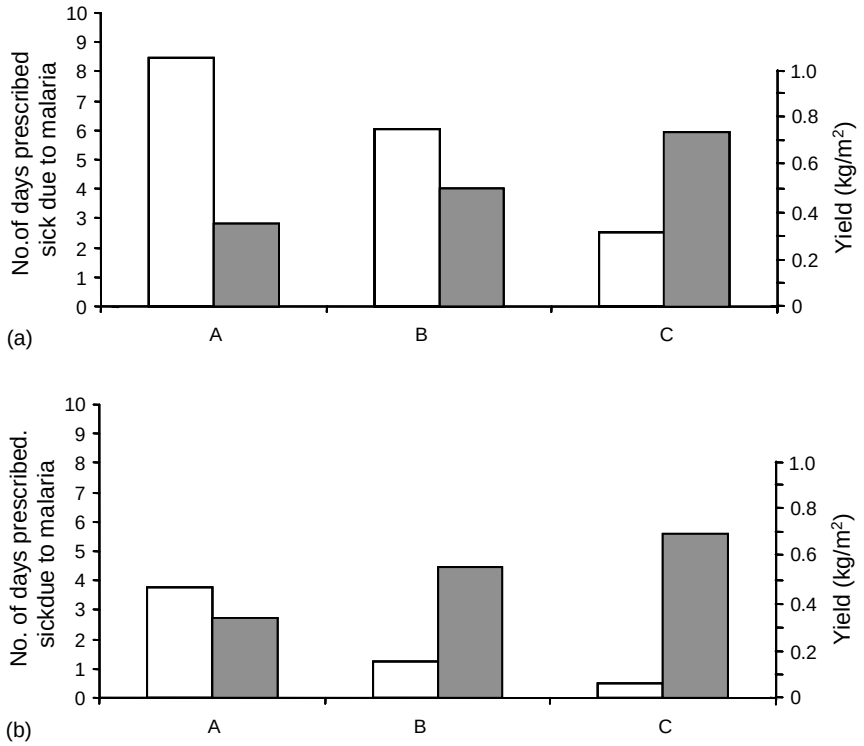


Fig. 4. Mean number of days farmers were classified as sick due to malaria (open bars) and mean yields (grey bars) stratified by three producer groups A, B and C for the okra cycle from September to December 2001 (a) and the subsequent cabbage cycle from January to June 2002 (b).

weeding and harvesting, but also 4 fertiliser applications, 11 phyto-sanitary treatments and daily watering in the absence of rain. Missing one of these steps can adversely affect overall productivity and result in the complete loss of production, and hence invested resources (both labour and money). Resource-use efficiency was sequentially increased when the project changed from the initial farmer-group production approach to individual plot management. This was due to a higher commitment of individual farmers. In the present study, most measures to control potential risk factors, which could inhibit successful vegetable farming, were made available by pre-financing, on-the-spot technical assistance, and application of successfully pre-tested cultivation itineraries. There was, however, one risk factor that could not be readily controlled, namely the health status of the participating farmers.

Besides reliance on external advice and support, and imported inputs, our findings also underscore that vegetable farming in the humid tropics is characterised by high labour intensity (CIRAD, 2002). Consequently, this production system is vulnerable to short durations of work absenteeism by farmers due to diseases or injuries. In fact, we found that when farmers were classified as sick for only a few days during critical periods of a single cabbage production cycle, it resulted in huge losses of yields and, consequently, revenues. In other words, the constant availability of a trained and healthy workforce to effectively carry out the necessary steps in a timely fashion during the entire cultivation cycle was of critical importance to guarantee high yields and hence income. This adds weight to the intimate connection between agriculture and health (Lipton and de Kadt, 1986; Mouchet and Brengues, 1990). Further, it confirms the economic impact of disease on farm-level economies, as observed also with other non-traditional agricultural products (Stanley, 2003).

Detailed analysis of routinely collected disease records from 1997–2001 revealed that malaria is the

main cause for health seeking in the local health care facilities, accounting for more than half of the total consultations. These findings were confirmed by the two cross-sectional surveys focussing on malaria parasite rates, primarily among school-age children, and the longitudinal assessment of infectiousness of malaria vectors. We found a mean EIR of 291 per year. *P. falciparum* was the predominant malaria parasite, transmitted by *A. gambiae s.l.* and *A. funestus*. Our findings confirm previous studies that malaria is hyper-endemic in the savannah zone of Côte d'Ivoire and that transmission is perennial (Dossou-Yovo et al., 2001; Audibert et al., 2003a, b; Henry et al., 2003; N'Goran et al., 2003). It is important to note that more than half of the interviewed household heads reported no particular means of protection against mosquito bites, hence malaria. Only 18% of the households reported the possession of bednets, few of which were impregnated with insecticides. Use of fumigating coils was employed by 11% of the households as the prime strategy for prevention of mosquito bites. In a recent study carried out in the northern part of Côte d'Ivoire, a similar low coverage of bednets was recorded, but much higher use of fumigating coils (De Plaen et al., 2003). Our findings also indicate that an improved house design, particularly the use of iron-corrugated roofing rather than thatched-grass, is negatively associated with mosquito bites. This might be explained by significant reductions of nuisance mosquitoes readily entering better constructed houses (Lindsay et al., 2002).

One of the most interesting findings in the present study was derived from the detailed health records that were kept on all 12 vegetable farmers over the course of the 10 months of farming, comprising two production cycles. On average, each farmer was classified as sick for 14–15 days due to diseases or injuries, with 8–9 days attributable to malaria (58%). It is important to note that such a high risk of malaria is expected in this epidemiological setting with stable transmission, hence is unlikely to be a consequence of the implementation of the farm-level drip irrigation system (Ijumba and Lindsay, 2001). In fact, well maintained drip irrigation is less associated with the transmission of water-related parasitic diseases when compared to traditional surface irrigation, due to the absence of open water bodies (Chimbari et al., 2004). In a comparable study carried out in a malaria-endemic area of Paraguay among communities that were primarily engaged in subsistence agriculture, malaria was responsible for 55% of all the time lost in the fields from diseases and injuries (Conly, 1972). Another more recent study carried out in a small-scale irrigation scheme in Zimbabwe found that malaria accounted for 38% of the total number of work days lost per annum (Boelee, 2003).

The strong heterogeneity of the total number of days farmers were classified as sick in the present study allowed comparison between two groups, stratified by a certain threshold of work absenteeism. For example, farmers who were classified as sick due to malaria for a maximum of 2 days gained twice the yield and twice the gross margin from their agricultural enterprises when compared to farmers with a higher number of days classified as sick. This finding underscores the great need to implement readily adapted measures to control malaria, in order to reduce farmers' vulnerability, which in turn will increase agricultural productivity and economic benefits (Fig. 1; loop R4). The study in Paraguay cited above provides supportive evidence, since the relative contribution of malaria to the rest of the diseases and injuries was reduced from 55 to 26% of absenteeism after implementation of sound control measures (Conly, 1972).

A host of previous studies have quantified the negative impact of malaria at the household level in societies that are primarily engaged in subsistence farming across diverse ecosystems (for a few selected examples, see Shepard et al., 1991; Mills, 1993; Konradsen et al., 1999; Attanayake et al., 2000). Moreover, it has been demonstrated that successful malaria control in industrial settings is a central feature in social and economic development with broad-based effects (Dua et al., 1997; Utzinger et al., 2002).

Finally, some important lessons can be extracted from the current study about how to improve community acceptance of development work more generally. A priori, the project was well received by the selected farmers, and also the broader community, since it raised expectations towards an agriculture-based rural development. However, because the initial support was allocated only to a small fraction of the community (Fig. 1; loops R1 and R2), other people in the village became exceedingly dissatisfied. Eventually this led to a land property conflict (Fig. 1; loop B1) and triggered an act of vandalism, resulting in

the partial destruction of the irrigation equipment. The farmers did not own the land they cultivated but, instead, the community lent it to them. Hence, it expected payoffs from the project. Resolution of this conflict was partially achieved when the project was linked with some other development work in the village, e.g. connection of the school to the water network, training of health personnel and teachers for early diagnosis and treatment of malaria with support from the national malaria control programme.

Since the inception of the project, half of the participating farmers have left the project. One farmer had to terminate production because of a hernia operation (Table 3; A4). Another farmer halted vegetable farming because he had accumulated sufficient funds to engage in an alternative and, in his own view, more interesting economic activity (Table 3; B4). Interestingly, the other 4 farmers who left (Table 3; A1, A2, A3 and B1) are those with the largest number of days classified as sick (19–32 days), primarily due to malaria (10–26 days). The remaining six farmers, who were less sick, continued with intensive vegetable farming without any subsidies. Their main challenge is to sustain sufficient liquidity for reinvestment in production inputs. This is partially due to high family expenses for food- and health-related issues (Fig. 1; loop B2). Six new farmers have been identified by the village authorities, exhibiting strong interest to join the project. Our findings suggest that it is imperative to substantially reduce the current burden of malaria in this epidemiological setting to maintain an incentive for farmers to persist with the intensive methods for vegetable farming. This could be achieved through a multi-stage process. First, personal protection (e.g. sleeping under insecticide-treated bednets) and improved house construction among farmers are likely to elevate their economic benefits. Second, more community-wide interventions should be implemented and sustained. This requires that enough families have higher economic resources, so that in concert with local governments and possibly outside donors, current malaria risk factors can be reduced on a broad scale. One might speculate that such a multi-stage process could substantially reduce malaria rates and in turn initiate a shift from subsistence agriculture to cash crops, as the route to getting local governments interested and committed towards malaria control.

In conclusion, a strong connection appears between intensive vegetable farming and the health status of farmers, which undoubtedly has an impact on the overall development of the communities concerned. The health impact is primarily due to malaria, which therefore exhibits high backward linkage, supporting recent claims that malaria and poverty are intimately connected (Sachs and Malaney, 2002; Utzinger et al., 2002). Although intensified production might be promising to break the cycle of poverty in systems of subsistence agriculture, health issues must be addressed simultaneously and cover far more than the direct targeted study. This calls for an integrated and inter-sectorial approach for mitigating the current burden of malaria, and suggests that malaria control interventions linked to agriculture might facilitate unprecedented and more equitable social and economic development in sub-Saharan Africa.

Acknowledgements

Our thanks are addressed to the village authorities of Tiémélékro and the 12 farmers and their families for their commitment to participate in the present study. We are indebted for financial support granted by the ‘Fonds Ivoiro Suisse de Développement Economique et Social’ (FISDES), the Swiss Academy of Sciences (SANW) and the Swiss Federal Commission for Fellowships for Foreign Students (CFBEE). This work has been partially supported by the National Centre of Competence in Research (NCCR) North–South “Research Partnerships for Mitigating Syndromes of Global Change”, Integrated Programme No. 4 (IP4) “Health and Well-being”. A. Tschannen acknowledges financial support by the Swiss Centre for International Agriculture, Swiss Federal Institute of Technology. J. Utzinger is grateful to the Center for Health and Well-being at Princeton University and the Swiss Tropical Institute for financial support.

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